

## 1. Background/context of the business scenario

This analysis explores data published by the NHS, focusing on the 11 months between August 2021 and June 2022, along with data scraped from Twitter for the same time period. The NHS datasets contain appointment-level records across GP practices in England, covering appointment status, mode, national category, and lead time.

It seeks to answer three key analytical questions:

1. Has there been adequate staff and capacity in the networks?
2. How are GP practices responding to demand pressure?
3. What can external data tell us that appointment records cannot?

## 2. Analytical approach:

### 2.1 Datasets

Four datasets were provided. Two were explored and excluded:

**Twitter:** sorted by hashtag, Healthcare dominates with 835 records. Closer analysis revealed that many comments are USA-based, making this dataset unreliable and irrelevant.

```
[112]: # Display records where the count is larger than 10.  
popular_hashtags = hashtag_counts[hashtag_counts['count'] > 10]  
popular_hashtags
```

```
[112]:
```

|     | hashtag                | count |
|-----|------------------------|-------|
| 0   | #healthcare            | 835   |
| 1   | #health                | 85    |
| 2   | #ai                    | 46    |
| 3   | #medicine              | 42    |
| 4   | #job                   | 38    |
| ... | ...                    | ...   |
| 44  | #innovation            | 11    |
| 45  | #usa                   | 11    |
| 46  | #digitaltransformation | 11    |
| 47  | #healthcareit          | 11    |
| 48  | #medtech               | 11    |

49 rows × 2 columns

**\*\*Observations:\*\*** Surprising to see AI and Jobs in the top 5 and disappointing not to see #NHS, #gpappointments, #gpwaitingtimes. This adds further weight to the hypothesis that the dataset will be of limited relevance to the analysis.

**actual\_duration (ad):** Excluded due to serious data quality limitations: duration recording methods vary by GP systems, and appointments with null, sub-one-minute, or over-60-minute durations are grouped into an 'Unknown/Data Quality Issue' category.

Both **national\_categories (nc)** and **appointments\_regional (ar)** were examined for nulls and duplicates. No nulls were found in either. Apparent duplicates were investigated and retained as valid multi-practice records. Each row represents aggregated appointment counts sharing identical attribute combinations.

## 2.2 Date Ranges and Granularity

The **nc** dataset covers 11 months (Aug 2021 to Jun 2022). The **ar** dataset covers 30 months but was filtered to the same time period for cross analysis.

| Dataset | From      | To        | Notes     |
|---------|-----------|-----------|-----------|
| ad      | 01-Dec-21 | 30-Jun-22 | 7 months  |
| ar      | Jan-20    | Jun-22    | 30 months |
| nc      | 02-Aug-21 | 30-Jun-22 | 11 months |

No data joins were possible as the datasets have different granularity levels and different date formats.

| Dataset | Row represents                                            |
|---------|-----------------------------------------------------------|
| ad      | Sub-ICB location name and code + date + duration band     |
| ar      | ICB (parent level) + month                                |
| nc      | Sub-ICB location name and code + date (incl time) + month |

## 2.3 Data Quality

Substantial data quality issues were identified in the metadata, meaning that all findings must be deemed indicative only:

- 13.1% of national categories are unmapped or inconsistently mapped
- 23.1% of lead times show as null
- Appointment counts are partially estimates
- Appointment mode is unreliable
- HCP type is only reliable for GP classification
- Appointment status has 3-6% monthly "Unknown" records.

## 2.4 Analytical Method

## NHS GP Appointments: Capacity and Access

Initial exploration started with the national picture then drilled down.

Chart 2: National capacity utilisation was calculated using the benchmark of 1.2M daily appointments and three progressively refined methods.

Chart 3: A day-of-week analysis of appointments confirmed the 5-day working week

Chart 4: ICB-level capacity was apportioned proportionally based on share of national appointments

Charts 5 & 6: Registered patient normalisation was applied for sub-ICB comparisons across the 5 largest ICBs by appointment. June 2022 data was used, matching the final month of the appointment period. NWL was identified as an outlier and Cheshire & Merseyside, with similar patient counts, was selected for comparison.

The April 2026 release of this data was initially trialled but rejected: registered patients grew by 2.2M between January 2022 and April 2026, with growth concentrated in urban areas, which would have distorted cross-ICB normalisation.

Line charts were used to show change over time, with different colours and line weights used to distinguish ICBs. Bar charts were used for Twitter hashtag analysis and for a categorical analysis of working days.

### 3. Visualisation and insights:

#### 3.1 Q1: Has there been adequate staff and capacity in the networks?

Chart 2

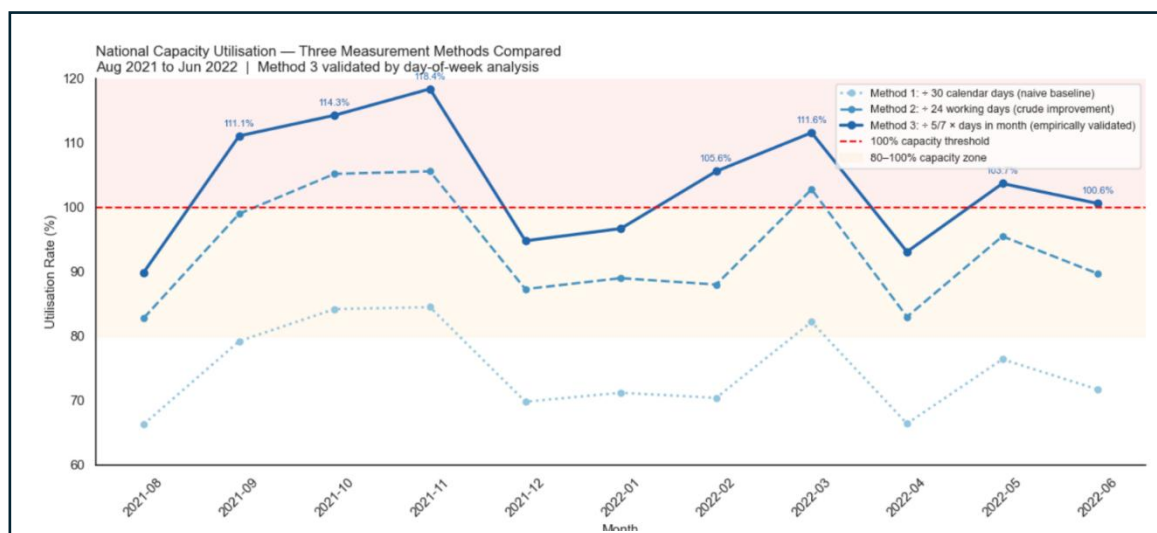


Chart 2 presents three distinct measures of capacity utilisation. Method 3 was tested and shown to be the most accurate (see Chart 3 below) and shows a system under pressure and frequently operating beyond maximum level over the 11-month period against the national benchmark.

Chart 3:

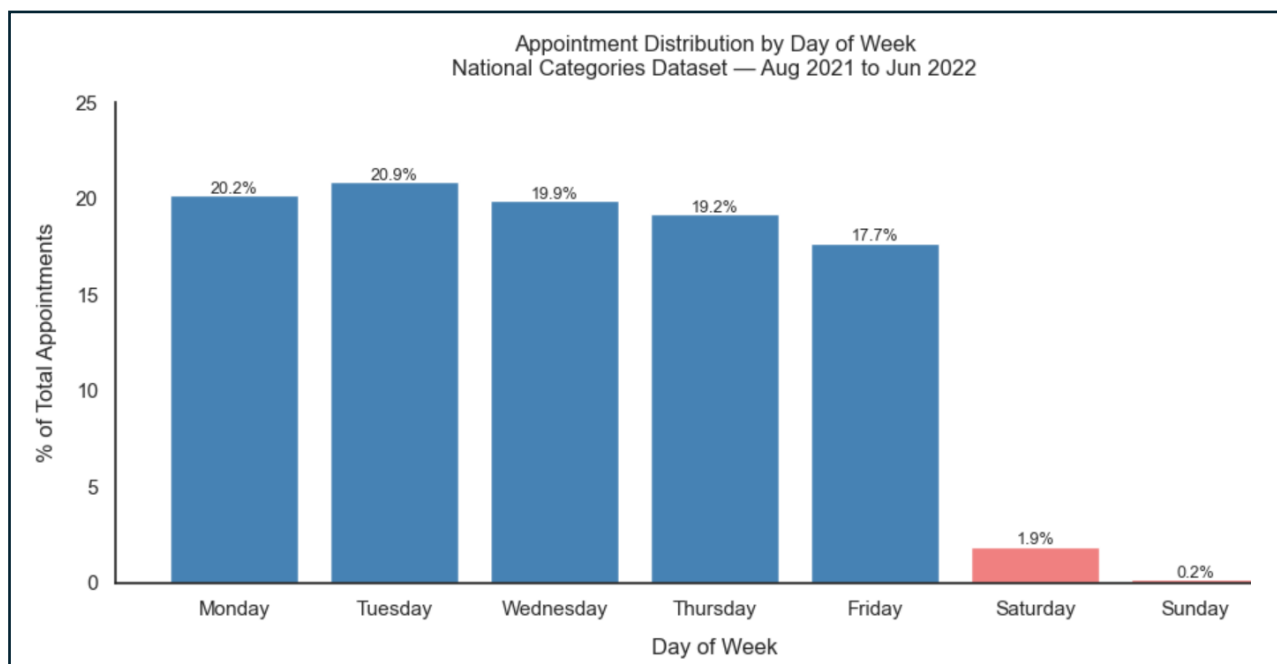


Chart 3 definitively validates the 5-day working week assumption. 97.9% of appointments are Monday-to-Friday and cell 193 confirms that Saturday appointments are rarely used for unplanned demand.

Analysis of HCP type shows a growth trend away from GPs towards Other Practice Staff over the period, consistent with ARRS, however data quality issues limit the conclusions that can be drawn from this finding.

### 3.2 Q2: How are GP practices responding to demand pressure?

Chart 4

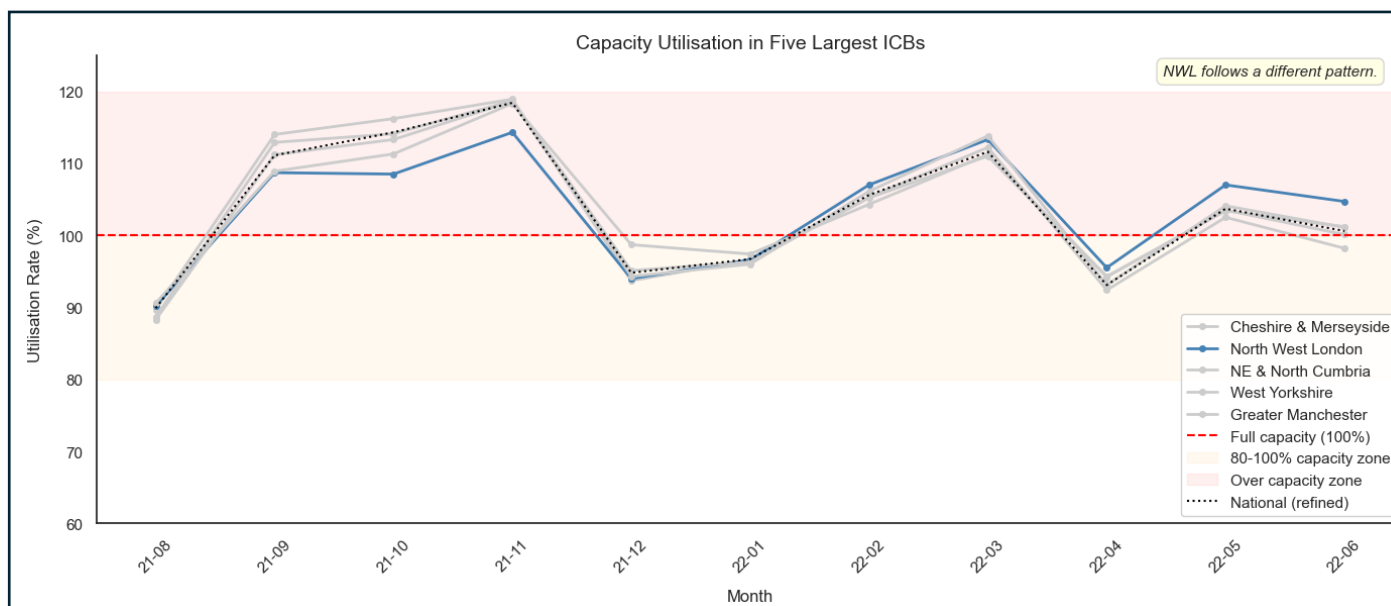
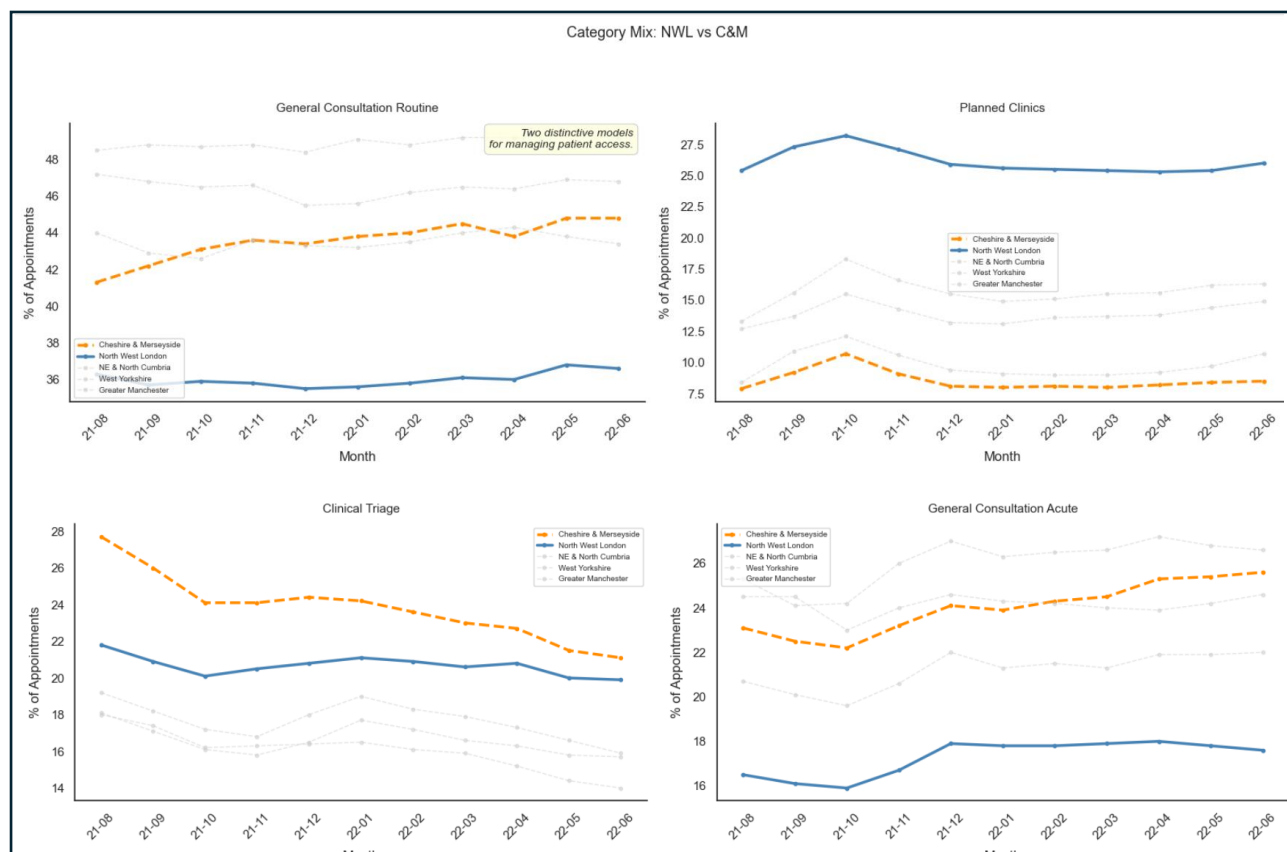


Chart 4 (refined method,  $\pm 5/7$  working days) shows NWL following a different trajectory from the national trend and other large ICBs. 6 hypotheses were tested and rejected before the key driver was identified in the category mix analysis.

Chart 5



## NHS GP Appointments: Capacity and Access

Chart 5 reveals 2 structurally different models of managing patient access:

NWL makes around three times greater use of planned clinics than the other large ICBs, with correspondingly lower rates of General Consultation Routine and Acute. This is a triage-first model which diverts routine and minor acute cases to free GP capacity for more complex cases.

C&M has the highest Clinical Triage rate (but this has been declining steadily) and also reports the highest General Consultation Routine rate. Here, triage functions as an appointment prioritisation tool - but almost all roads lead to a GP.

Chart 6

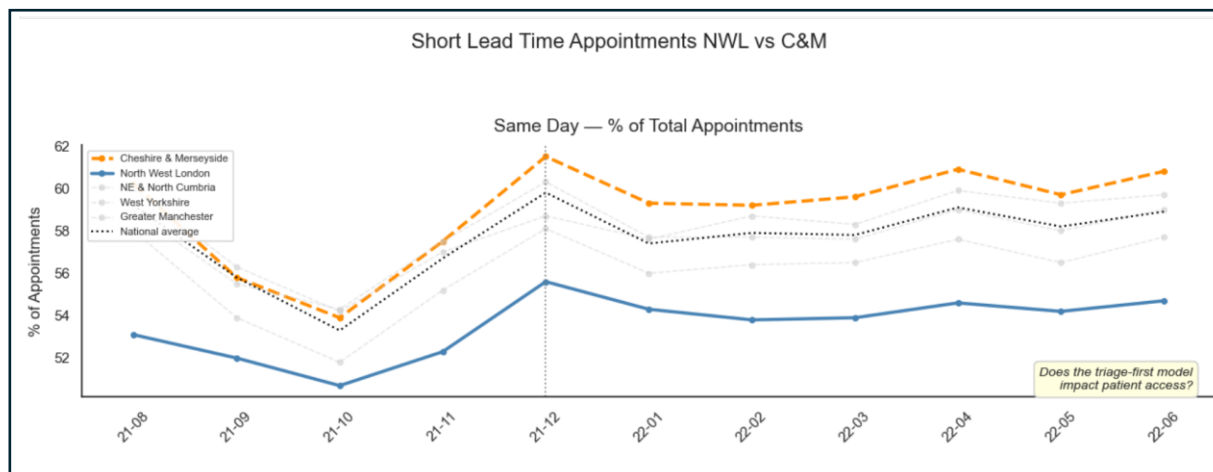
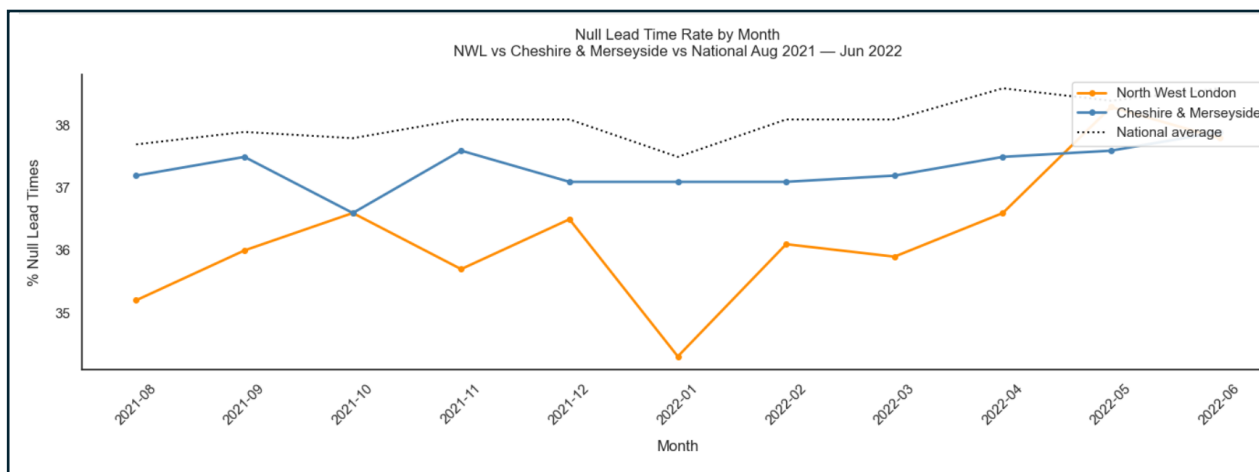


Chart 6 reveals a paradox: NWL consistently has the lowest same-day appointment rate while C&M has the highest. NWL's lead-time profile prioritises efficient allocation of capacity over immediate access. The below check on null lead-times showed NWL's to be low. This finding is not the result of poor recording and should be treated as genuine.

Chart 6b (not included in the presentation)



**DNA rates** nationally are low (4-5.5%) and relatively stable, suggesting missed appointments are not a primary driver of capacity pressure. NWL's mean rate is 4.8%, placing it 14<sup>th</sup> out of 42 ICBs. This is above average but not in the top quartile.

### 3.3 Q3: What can external data tell us?

The Twitter dataset was excluded (see above). Web scraping is notoriously unreliable, however a recommendation is made to survey patients directly, using AI to automate the process.

## Patterns and predictions:

Using the empirically validated measure of a 5-day working week and the published 1.2M daily appointments benchmark, the NHS operated nationally above 100% capacity for 7 of 11 months.

Capacity pressure peaks in autumn and March, with dips in December and April. Given the post-COVID context here, these patterns cannot be assumed to be typical.

DNA rates are low and stable nationally (4-5.5%) and are not a primary driver of capacity pressure.

NWL's triage-first model appears to manage GP workload effectively but restricts same-day access. C&M offer better same-day access but their triage model routed very few patients away from GPs and its use has steadily declined. Neither model is optimal: NWL sacrifices access for efficiency, C&M sacrifices efficiency for access.

The time between booking and appointment is recorded in unequal bands, with no distinction between waiting 2 days and waiting 7 days for an unplanned illness. The published data therefore cannot support clinical safety monitoring of short waits. More granular bands would close that gap, particularly when evaluating triage-first models such as NWL's, where same-day access is demonstrably restricted.

Better data does not increase capacity, but it enables the same capacity to be used more effectively, which, in a system consistently operating at or beyond capacity, as this analysis has shown, represents a substantial opportunity.

# Appendices

## References

NHS Digital, *Patients Registered at a GP Practice, June 2022 (Totals and Mapping files)*. Available at: <https://digital.nhs.uk/data-and-information/publications/statistical/patients-registered-at-a-gp-practice/june-2022> (accessed 7.5.26)

 **Patients Registered at a GP Practice – June 2022: Totals (GP practice-all persons)** ↓

CSV 481 KB

 **Patients Registered at a GP Practice – June 2022: Mapping (Commissioning Regions- STPs-CCGs-PCNs- GP practice)** ↓

CSV 1 MB

Bakhai M et al. *How to align capacity with demand in general practice*. NHS England, 2024 (updated October 2025). Available at: <https://www.england.nhs.uk/long-read/how-to-align-capacity-with-demand-in-general-practice/>

NHS England, *Modern General Practice Model* <https://www.england.nhs.uk/gp/national-general-practice-improvement-programme/modern-general-practice-model/> accessed 16.5.26

NHS England, *Recording same day appointments for all clinically urgent patients* <https://www.england.nhs.uk/long-read/recording-same-day-appointments-for-all-clinically-urgent-patients/> accessed 18.5.26

*Storytelling with Data: A Data Visualization Guide for Business Professionals*, Knafllic, 2015

*Storytelling with Data: Before & After*, Knafllic 2025

NHS England, *Expanding our Workforce (ARRS reference)*:

<https://www.england.nhs.uk/gp/expanding-our-workforce/> accessed 4.5.26



## Notes on Layout of Jupyter Notebook

The notebook in this repository is an extract, restructured into six sections following the analysis rather than the order in which the work was carried out: datasets and exclusions, national capacity, ICB-level analysis, category mix, lead times and summary. Its section numbering is independent of the numbering used in this report.

## Use of AI

AI assistance (Claude, Anthropic) was used in this project for the following purposes:

1. Explanation of Python concepts and syntax
2. Troubleshooting of technical errors during data import and query execution
3. Production of code snippets based on my analytical direction and explanation of these in plain English

All analytical decisions, business interpretations, query objectives, and findings are my own work and I am able to explain and defend all technical content.